Author Contribution Statement

Field Validation of the Self-Adaptive Model-Based Protection System on 157 Production IBR Fault Recordings from a 33 kV DSO Network

CRediT Taxonomy

Contribution	A. V. Eluvathingal	R. K. Prasad
Conceptualisation	Lead	Supporting
Data curation	Lead	—
Formal analysis	Lead	Supporting
Funding acquisition	—	Lead
Investigation	Lead	Supporting
Methodology	Lead	Supporting
Project administration	—	Lead
Software	Lead	—
Supervision	—	Lead
Validation	Lead	Supporting
Visualisation	Lead	—
Writing — original draft	Lead	—
Writing — review & editing	Supporting	Lead

Plain-Language Statement (for portal upload)

Arun V. Eluvathingal: Conceptualised the field-validation framework; curated and labelled the 157-recording DSO-A dataset (TR-73); developed the three field-adaptation corrections F1 (legacy-IBR κ_n-trajectory detector), F2 (split-window HIF estimator), and F3 (shape-factor calibration); performed all analyses and Monte Carlo benchmarks; produced all figures; wrote the manuscript draft.

R. K. Prasad: Supervised the research programme; secured the research agreement with DSO-A for field-data access; provided guidance on experimental design and result interpretation; reviewed and edited the final manuscript.

Conflict of interest: The authors declare no conflict of interest. The field dataset was provided under a research agreement with a distribution network operator; the operator had no role in analysis, interpretation, or the decision to publish.

Data availability: Anonymised event metadata (timestamps, fault type, $\mathrm{k}_{\mathrm{ibr}}$ quartile) will be made available via the IIT Madras data repository upon acceptance. Raw DFR waveforms cannot be shared publicly due to the DSO confidentiality agreement.